Lesson 1: Linear Search

Module 1 | 45 minutes | Searching algorithms | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. linear search ___ A. The input that makes an algorithm do the most work.
b. target ___ B. Checking if two values are equal.
c. comparison ___ C. The value being searched for.
d. worst case ___ D. Checking items one by one in order.

2. Plan your code

Before you build it: Find a friend: names is filled. Ask “Who are you looking for?”. Search the list; if found, say “Found at position ” joined with the position; otherwise say “Not found”. Write or draw the blocks you will use.

3. Worst case

A list has 1,000 items. In the WORST case, how many items does linear search check? Circle one.

a. A. 1,000
b. B. 500
c. C. 10

4. Plan your code

Before you build it: Count the work: search names for “Zoe” (no ask needed) and count each comparison in steps. Zoe is item 5, so steps should be 5. Write or draw the blocks you will use.

5. Search in Python

What does this print? (Python counts from 0.) Code: def find(items, target): ↵ for i in range(len(items)): ↵ if items[i] == target: ↵ return i ↵ return -1 ↵ ↵ print(find(["a", "b", "c"], "c"))

a. A. 2
b. B. 3
c. C. -1

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can implement linear search.
☐ I can handle the not-found case.
☐ I can measure an algorithm’s steps.